

Progression Outline

Complete Rule Table (In order of introduction)		
Accessible name	“Professional” name	Action
Yanking	Yanking / Snake equations	“Shake” an endpoint
Spider Fusion	Spider Fusion	Drag two spiders together
Identity	Identity	
Leg Chop	Hopf	
Copy	Copy	
Square Popping	Bialgebra	

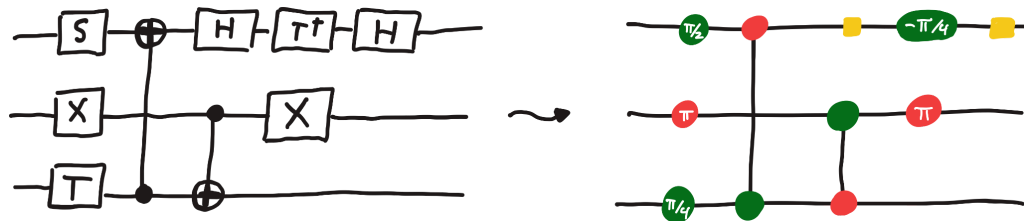
General Comments/Questions:

- We should make clear that all of these rules are two-sided

Note: the following is subject to change,

0. (General Introduction)

“All Quantum circuits are made of spiders and wires...”



Left: Quantum circuit written with unitary boxes

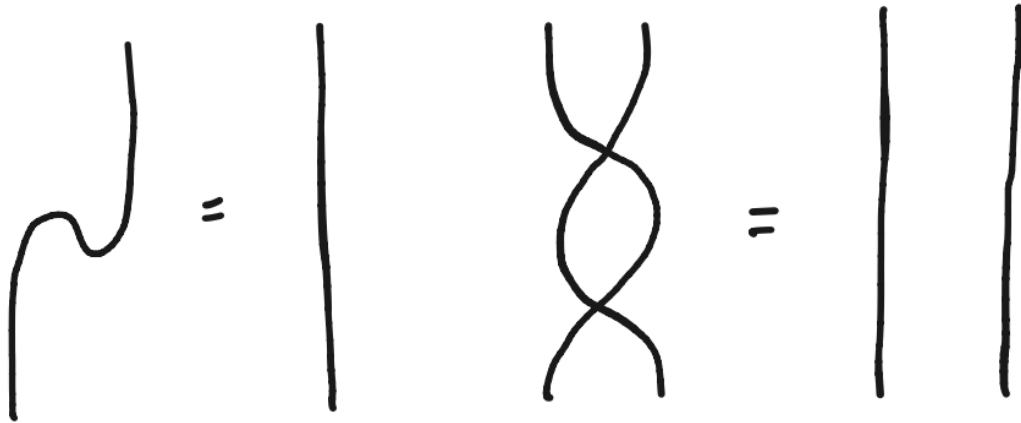
Right: The same circuit written with spiders and wires!
(those yellow boxes are also spiders, as we will soon see)

Source: <https://arxiv.org/pdf/2012.13966>

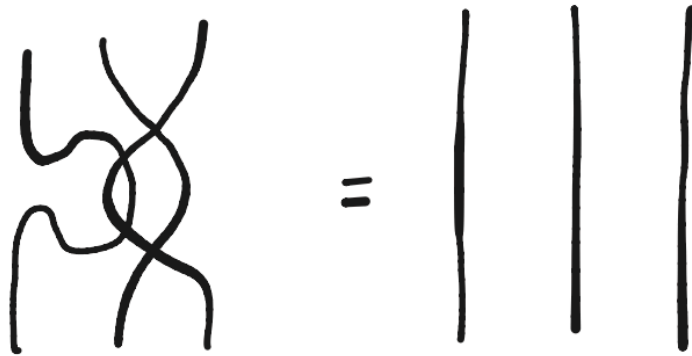
“This fact lets us reason about quantum circuits using the ZX calculus.”

1. (Yanking)

“Sometimes wires get a little disorganized. Like normal wires, only their inputs and outputs are what matter, so we can always straighten them out to make a diagram nicer”

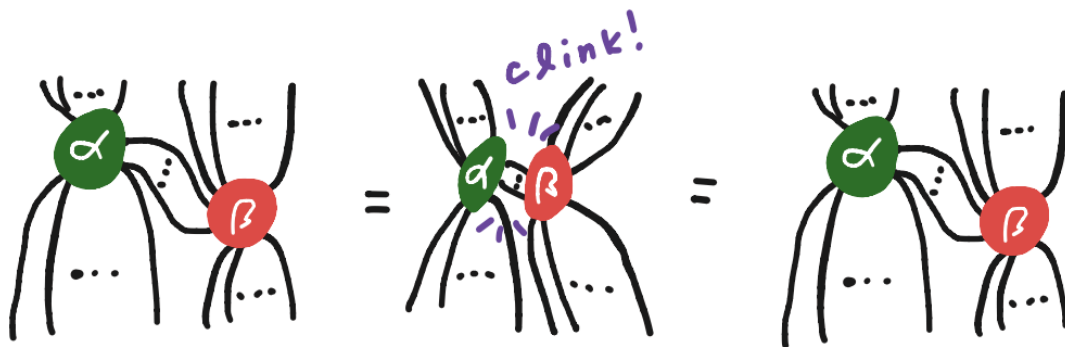
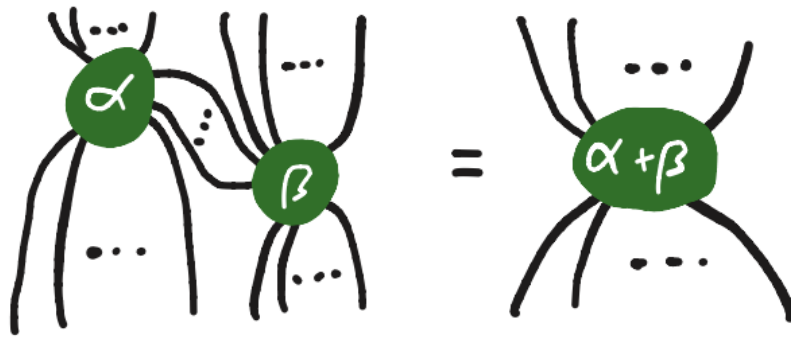


Exercise: “Shake the wires to straighten them!”

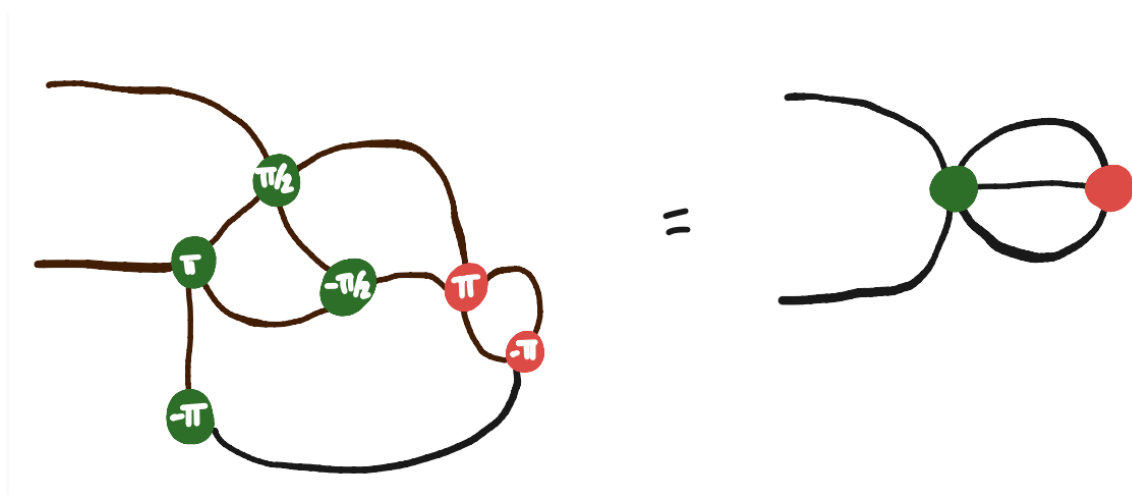


2. (Spider Fusion)

“Spiders of the same color can fuse into each other if they’re connected by a plain wire. Opposite colors do not.”

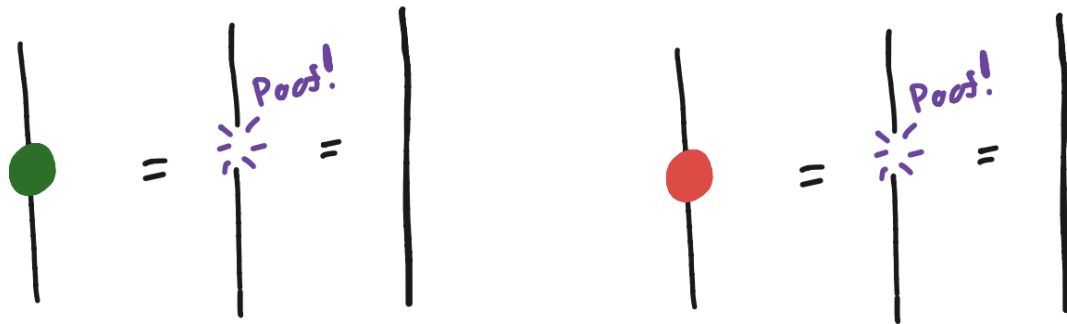


Exercise: "Simplify the following circuit using spider fusion!"

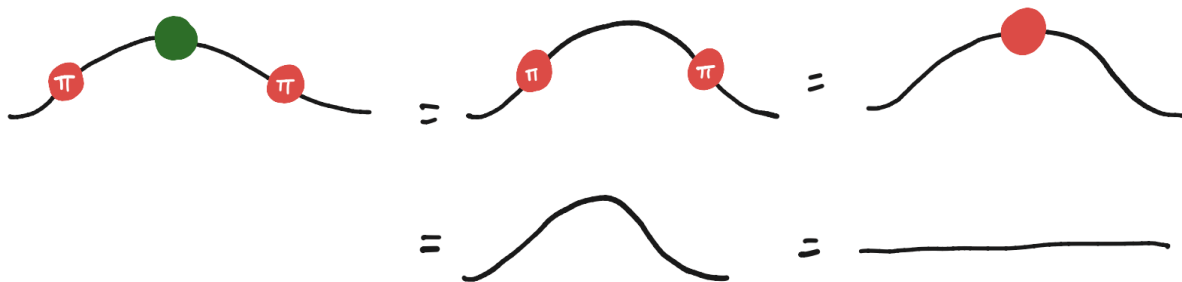


3. (Identity)

"When a phaseless spider only has one input and one out, you can remove it from the wire"

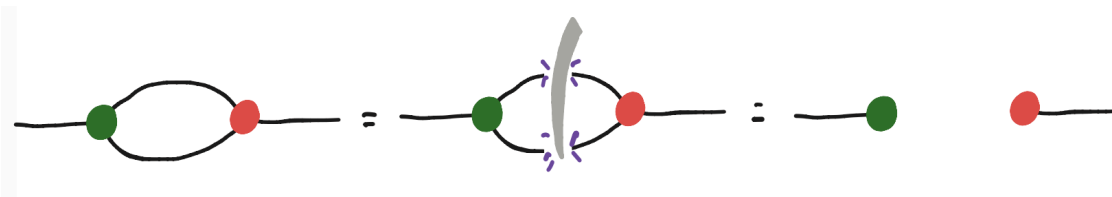


Exercise: "Use identity removal to turn the following into an identity wire..."



4. (Leg Chop)

"When a green spider is connected to a red spider by two parallel wires, you can 'chop' them off. (sorry, spiders)"

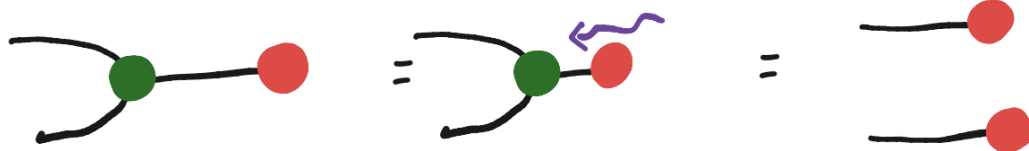


Exercise: "Use leg chopping to simplify the following diagram"

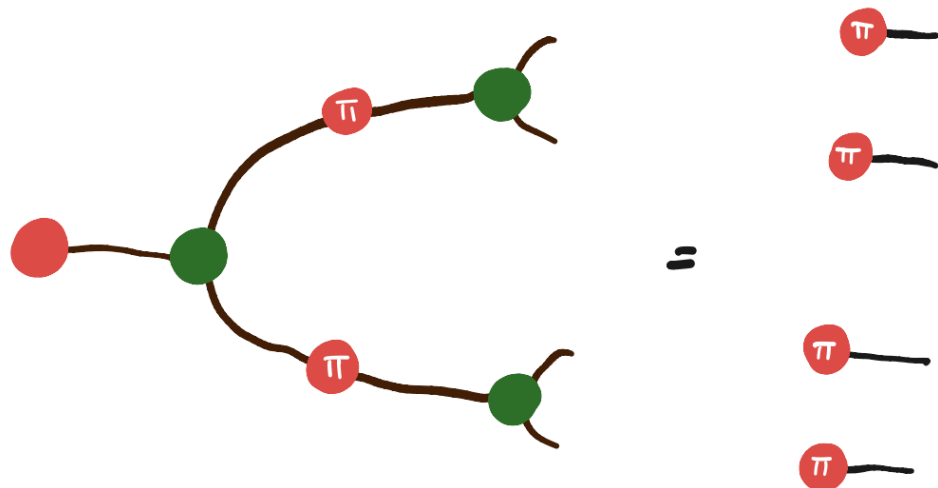


5. (Copy)

“We can push blank (or phase π) red spiders through any blank green spider in the following way. We’re basically copying the red spider across all the wires connected to the green spider.”

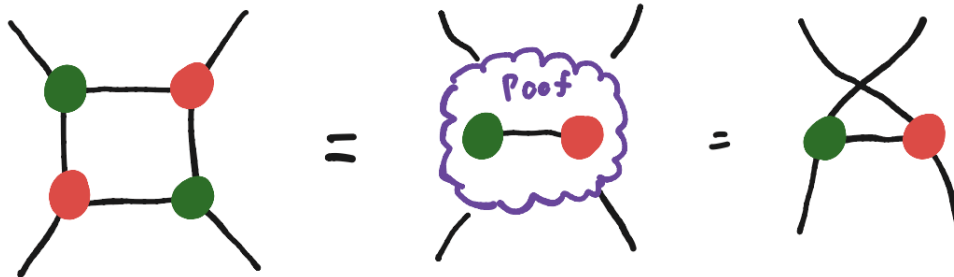


Exercise: “Show that we can use this to simplify the following diagram”

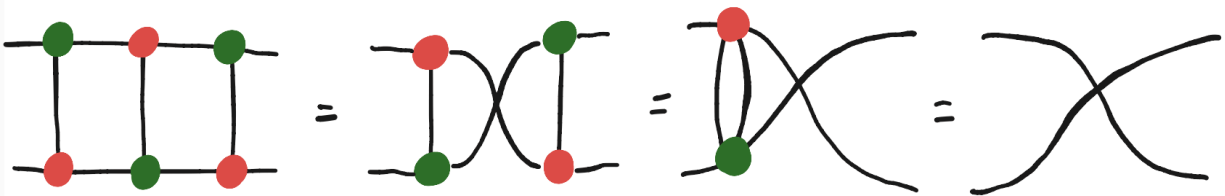


6. (Square Popping)

“Now for one of the most useful rewrites -- ‘square popping.’ Given a square of spiders in the following setup, you can simplify it to just two.”



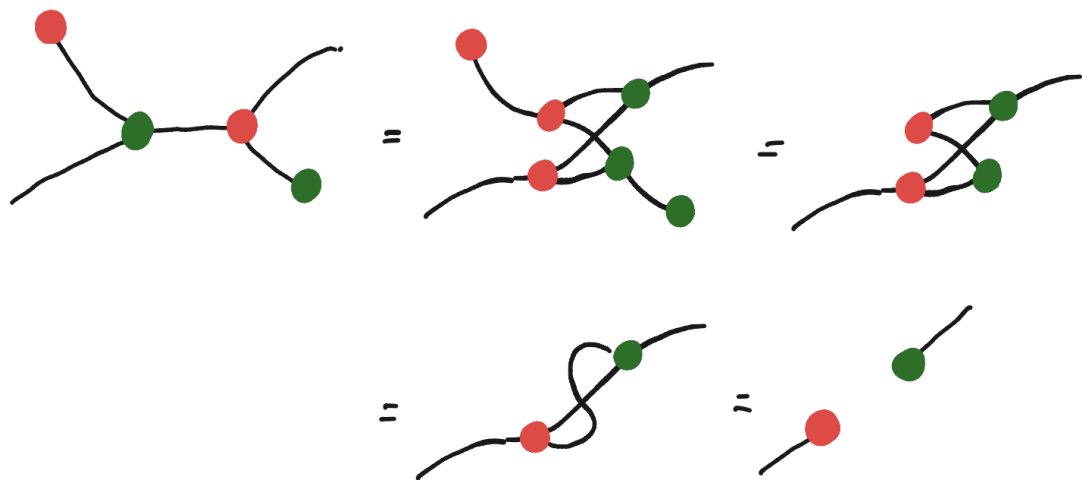
Exercise: “Let’s use this powerful rewrite to simplify the diagram:”



“Conversely, we can use this rule to ‘push’ a green spider through a red spider:”



Exercise: “Push the green spider through the red one to simplify the following diagram” [Disable copy]



7. (Final Exercise)

Exercise: "Prove that Square Popping lets us leg chop!" [Disable hopf]

